

EX. NO : 2a

Tic Tac Toe Game

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Aim:

To develop a console-based 2-player Tic Tac Toe game in Python, where two players alternate turns to place their marks (X and O) on a 3x3 board, and the program determines the winner or a draw.

Procedure:

1. Initialize the board:

Create a 3x3 matrix (list of lists) filled with empty spaces " ".

2. Display the board:

Implement a function to print the current state of the board after every move, showing the grid clearly.

3. Take input from players:

Alternate between Player X and Player O, asking for row and column inputs (0 to 2).

4. Validate moves:

Check if the selected cell is empty; if not, prompt the player to try again.

5. Update the board:

Place the player's mark in the selected cell.

6. Check for a win:

After each move, check rows, columns, and diagonals for three identical marks.

7. Declare results:

- If a player wins, display the winner and end the game.
- If all 9 moves are made without a winner, declare the game a draw.

Program Coding:

```
def print_board(board):
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    for row in board:
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        print(" | ".join(row))
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    print("-" * 5)

def check_win(board, player):

    # Rows, columns, diagonals

    for i in range(3):

        if all([cell == player for cell in board[i]]): return True

        if all([board[j][i] == player for j in range(3)]): return True

    if all([board[i][i] == player for i in range(3)]): return True

    if all([board[i][2 - i] == player for i in range(3)]): return True

    return False

def play_game():

    board = [[" " for _ in range(3)] for _ in range(3)]

    current = "X"

    for turn in range(9):

        print_board(board)

        row = int(input(f"Player {current}, enter row (0-2): "))

        col = int(input(f"Player {current}, enter col (0-2): "))

        if board[row][col] != " ":

            print("Cell taken! Try again.")

            continue

        board[row][col] = current

        if check_win(board, current):

            print_board(board)

```

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        print(f'Player {current} wins!')

    return

    current = "O" if current == "X" else "X"

    print_board(board)

    print("It's a draw!")

play_game()

```

Output:

- ☐ A 3×3 board is created.
 - ☐ Two players ("X" and "O") take turns.
 - ☐ The game checks for a win or draw after each move.
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Player X, enter row (0-2): 0

Player X, enter col (0-2): 0

X | |

| |

| |

Player O, enter row (0-2): 1

Player O, enter col (0-2): 1

X | |

| O |

| |

Player X, enter row (0-2): 0

Player X, enter col (0-2): 1

X | X |

| O |

| |

Player O, enter row (0-2): 2

Player O, enter col (0-2): 2

X | X |

| O |

| | O

Player X, enter row (0-2): 0

Player X, enter col (0-2): 2

X | X | X

| O |

| | O

✓ Player X wins!

Result:

Thus, the Tic Tac Toe game program was successfully implemented